

Division in the Integers

Let a be a nonzero integer and let b be an integer. We say that a divides b if and only if there is an integer c such that $b = ac$. If a divides b , then we use the notation:

$$a|b.$$

If a does not divide b , then we use the notation:

$$a \nmid b.$$

When a divides b , we say a is a divisor of b and b is a multiple of a .

Some useful facts about divisibility

- If $d|m$ and $d|n$, then $d|(m+n)$.

Proof: Let $m = ad$ and $n = bd$, then $m+n = ad+bd$
 $\Rightarrow m+n = d(a+b) \Rightarrow d|(m+n).$

- If $d|n$ and $n \neq 0$, then $d \leq n$.

Proof: Let $d|n$ then we have $n = kd \neq 0$

$$\Rightarrow k \geq 1$$

$$\Rightarrow n = kd \geq d.$$

- For all d , $d|0$.

Proof: $0 \cdot d = 0$

- If $d|m$ or $d|n$, then $d|mn$

Proof: If $d|m \Rightarrow m = kd \Rightarrow mn = (kn)d \Rightarrow d|mn.$

- If p is prime, then $p|ab$ if and only if $p|a$ or $p|b$.

- If $a|b$ and $b|c$ then $a|c$.

Proof: $b = k_1 a$ and $c = k_2 b \Rightarrow c = k_1 k_2 a \Rightarrow a|c.$

Division Algorithm

Let a be an integer and d be a positive integer. Then, there are unique integers q and r , with $0 \leq r < d$, such that $a = dq + r$.

Let a be an integer and d be a positive integer. Suppose that q and r are integers given by the division algorithm. We refer to q as the quotient and r as the remainder. We use the following notation:

$$\boxed{\begin{array}{l} a \text{ div } d = q, \\ a \text{ mod } d = r. \end{array}}$$

Example Find $129 \text{ div } 7$ and $129 \text{ mod } 7$.

Solution: We have

$$129 = 7 \cdot 18 + 3$$

$$\Rightarrow 129 \text{ div } 7 = 18$$

$$\text{and } 129 \text{ mod } 7 = 3.$$

Theorem: Let a be a positive integer and let b be an integer. Then $a \mid b$ if and only if $b \text{ mod } a = 0$.

For any positive integer m , we define Z_m to be the set of nonnegative integers less than m . In other words, we have

$$Z_m = \{0, 1, 2, 3, \dots, m-1\}$$

$$\text{Note: (i) } a +_m b = (a + b) \text{ mod } m$$

$$(ii) \ a \cdot_m b = (a \cdot b) \text{ mod } m$$

Theorem: Every positive integer $n > 1$ can be written uniquely as $p_1^{k_1} p_2^{k_2} p_3^{k_3} \dots p_s^{k_s}$, where $p_1 < p_2 < p_3 < \dots < p_s$ are distinct primes that divide n and the k_i 's are positive integers giving the number of times each prime occurs as a factor of n .

Example: (a) $9 = 3 \cdot 3 = 3^2$

(b) $24 = 8 \cdot 3 = 2 \cdot 2 \cdot 2 \cdot 3 = 2^3 \cdot 3$

(c) $30 = 2 \cdot 3 \cdot 5$

Greatest Common Divisor and Least Common Multiple

(i) Let a and b be positive integers. The largest positive integer d such that d divides a and d divides b is referred to as the greatest common divisor of a and b . The greatest common divisor of a and b is denoted by $\gcd(a, b)$.

If $\gcd(a, b) = 1$, we say that a and b are relatively prime.

Example: Find the greatest common divisor of 35 and 21.

Are 35 and 21 relatively prime?

Solution: Since $35 = 7 \cdot 5$ and $21 = 7 \cdot 3$, we see that 7 is a common divisor of 35 and 21. It can be checked that no positive integer larger than 7 that divides both 35 and 21. Thus we see that

$$\gcd(35, 21) = 7.$$

Since $\gcd(35, 21) \neq 1$, we see that 35 and 21 are not relatively prime.

Additionally, $\gcd(5, 3) = 1$, so $\frac{35}{7}$ and $\frac{21}{7}$ are relatively prime. //

(ii) let a and b be positive integers. The smallest positive integer m such that a divides m and b divides m is referred to as the least common multiple of a and b . The least common multiple of a and b is denoted by $\text{lcm}(a, b)$.

Example: Find the least common multiple of 35 and 21.

Solution: Since $105 = 35 \cdot 3$ and $105 = 21 \cdot 5$, we see that 105 is a common multiple of 35 and 21. It can be checked that no positive integer smaller than 105 is a multiple of both 35 and 21. Thus, we see that $\text{lcm}(35, 21) = 105$.

Additionally, we observe that $\text{gcd}(35, 21) \cdot \text{lcm}(35, 21) = 7 \cdot 105 = 735 = 35 \cdot 21$.

In general, we have the following theorem.

Theorem: If a and b are two positive integers, then $\text{gcd}(a, b) \cdot \text{lcm}(a, b) = a \cdot b$.

Example: let $a = 540$ and $b = 504$.

Factoring a and b into primes, we get

$$a = 540 = 2^2 \cdot 3^3 \cdot 5$$

$$b = 504 = 2^3 \cdot 3^2 \cdot 7$$

which can be written as

$$540 = 2^2 \cdot 3^3 \cdot 5^1 \cdot 7^0 \text{ and } 504 = 2^3 \cdot 3^2 \cdot 5^0 \cdot 7^1$$

$$\begin{aligned} \text{Now, } \text{gcd}(540, 504) &= 2^{\min(2,3)} \cdot 3^{\min(3,2)} \cdot 5^{\min(1,0)} \cdot 7^{\min(0,1)} \\ &= 2^2 \cdot 3^2 \cdot 5^0 \cdot 7^0 = 36 \end{aligned}$$

$$\text{and } \text{lcm}(540, 504) = 2^{\max(2,3)} \cdot 3^{\max(3,2)} \cdot 5^{\max(1,0)} \cdot 7^{\max(0,1)} \\ = 2^3 \cdot 3^3 \cdot 5^1 \cdot 7^1 = 7560$$

$$\text{Therefore, } \text{gcd}(540, 504) \cdot \text{lcm}(540, 504) \\ = 36 \cdot 7560 = 540 \cdot 504 //$$

Theorem: If d is $\text{gcd}(a, b)$, then

- (a) $d = sa + tb$, for some integers s and t .
- (b) If c is any other common divisor of a and b , then $c|d$.

Proof: Let x be the smallest positive integer such that $x = sa + tb$, where s and t be some integers.

Also let c be the common divisor of a and b , then $c|a$ and $c|b$.

It follows that $c|sa + tb \Rightarrow c|x \Rightarrow c \leq x$.

Now, we shall show x is a common divisor of a and b .

We know, $a = qx + r$ with $0 \leq r < x$

$$\text{or, } r = a - qx = a - q(sa + tb)$$

$$= a - qsa - qtb = (1 - qs)a + (-qt)b$$

If r is not zero, then since $r < x$ and r is the sum of a multiple of a and multiple of b , which contradicts to a fact that x is the smallest positive integer that is a sum of multiples of a and b . Thus r must be 0 and $x|a$.

In the same way, we can show that $x|b$ and this completes the proof.

Example: (a) The common divisors of 12 and 30 are 1, 2, 3 and 6.

$$\gcd(12, 30) = 6 \text{ and } 6 = 1 \cdot 30 + (-2) \cdot 12$$

$$(b) \gcd(17, 95) = 1 \text{ and } 1 = 28 \cdot 17 + (-5) \cdot 95$$

We now present a procedure, called the Euclidean algorithm, for finding $\gcd(a, b)$ conveniently. (By Example)

Example: Compute $\gcd(273, 98)$ using the Euclidean algorithm.

Solution:

$$\text{Divide } 273 \text{ by } 98: \quad 273 = 2 \cdot 98 + 77$$

$$\text{Divide } 98 \text{ by } 77: \quad 98 = 1 \cdot 77 + 21$$

$$\text{Divide } 77 \text{ by } 21: \quad 77 = 3 \cdot 21 + 14$$

$$\text{Divide } 21 \text{ by } 14: \quad 21 = 1 \cdot 14 + 7$$

$$\text{Divide } 14 \text{ by } 7: \quad 14 = 2 \cdot 7 + 0$$

So $\gcd(273, 98) = 7$, the last of the nonzero divisors. //

Que: If $\gcd(273, 98) = 7$, then express $\gcd = sa + tb$.

Solution: $\gcd(273, 98) = 7$

$$= 21 - 1 \cdot 14$$

$$= 21 - 1(77 - 3 \cdot 21)$$

$$= 4 \cdot 21 - 1 \cdot 77$$

$$= 4(98 - 1 \cdot 77) - 1 \cdot 77$$

$$= 4 \cdot 98 - 4 \cdot 77 - 1 \cdot 77$$

$$= 4 \cdot 98 - 5 \cdot 77$$

$$= 4 \cdot 98 - 5(273 - 2 \cdot 98)$$

$$= 14 \cdot 98 - 5 \cdot 273$$

$$\text{Where } s = -5 \quad t = 14 //$$